

Harshad Patel

harshad.hmt.mf@gmail.com
Gandhinagar, Gujarat, India
9016214369

Open to Remote, Open to Relocate, Open to Hybrid, Open to Travel, Open to On-Site
[LinkedIn](#), [Harshad Portfolio](#)

I believe technology creates real value when it solves real problems, works reliably in the real world, and supports measurable business outcomes.

I'm an Electronics Engineer, Embedded Systems Architect, Technology Leader, and Entrepreneur with 24+ years of experience across embedded systems, industrial automation, IoT/IIoT, electronics, networking, security, access control, surveillance, SCADA, control systems, and connected products.

Throughout my career, I have worked across the complete technology lifecycle—from understanding business and customer requirements to system architecture, hardware and firmware development, industrial communication, software integration, testing, deployment, troubleshooting, and customer delivery.

My strength is connecting technology, engineering, people, and business.

• Embedded Systems & Electronics • Industrial Automation & Control Systems • IoT, IIoT & Connected Devices • PLC, SCADA, Modbus & Industrial Communication • Hardware–Firmware–Software Integration • System & Product Architecture • Automation & Digital Transformation • Product Development & Engineering Strategy • Technical Project & Team Leadership • Technology Consulting & Customer Solutions As CEO at Dazzle Technologies, I am focused on building practical technology solutions that bring together embedded electronics, automation, IoT, software, data, and AI to address real industrial and business challenges.

I enjoy taking complex ideas and turning them into structured, practical, and scalable solutions—while balancing technical feasibility, cost, reliability, timelines, and business objectives.

My career is now focused on opportunities where I can contribute beyond individual engineering tasks—as a Technology Leader, Systems Architect, Product Builder, Technical Manager, Project Leader, or Strategic Technology Consultant.

I am particularly interested in organizations working on Embedded Systems, Industrial Automation, IoT/IIoT, Electronics, AI-enabled industrial solutions, smart products, and digital transformation.

→ Define the right architecture → Build with engineering discipline → Deliver reliably → Continuously improve.

I'm open to connecting with founders, CXOs, technology leaders, engineering teams, manufacturers, investors, and organizations building ambitious technology products and industrial solutions.

If you are looking for someone who can understand both the technical depth and the bigger business picture, I would be happy to connect.

Work Experience

CEO & Embedded System Architect • Dazzle Technologies • Himatnagar, Gujarat, India Sep '09 – Present

Spearheaded end-to-end product development for 20+ embedded systems across IoT, industrial automation, agriculture, and government projects, delivering energy-conservation solutions for water supply, street lighting, and remote monitoring

- Architected and launched 12+ custom products including PLCs, IoT gateways, BMS units, and remote terminal units (RTUs), achieving 30% faster deployment through modular firmware design
- Led cross-functional teams of 3–6 engineers; reduced prototype-to-production cycle by 25% by streamlining hardware-software integration and testing
- Directed ISO 9001:2015 certification compliance, authoring full quality documentation and audit procedures
- Engineered MODBUS RTU/TCP, MQTT, and GSM-based communication protocols, enabling real-time remote monitoring for 50+ client field installations
- Managed full product lifecycle: market research
- client requirement analysis
- schematic/PCB design
- firmware coding (C/C++, Assembly)
- prototyping
- field testing
- release
- Key Projects: PLC for Cold Storage: Designed LPC1833-based PLC with FreeRTOS; integrated MODBUS and HMI/SCADA interfaces; reduced manual temperature monitoring by 80%
- BMS for Automotive (OEM): Developed STM32-based battery management system communicating via CAN bus; achieved 99.9% data accuracy in battery health reporting
- IoT Gateway: Built ESP32-based gateway with MQTT/WiFi; aggregated data from 100+ field devices; enabled cloud-based dashboard for real-time analytics

- Remote Monitoring Unit (RMU): Created P89C669-based RMU for cross-country pipeline cathodic protection; integrated GPS-synchronized logging with 1-sec to 24-hour intervals
- Vehicle Tracking System (VTS): Delivered ARM7-based VTU with GPS/GSM; supported 10,000 waypoints and event-driven SMS alerts; reduced fleet response time by 40%.

Team Leader & Sr. Embedded Programmer • CIMCON Software, Inc. • Ahmedabad, Gujarat, India Sep '06 – Aug '09

Led a team of 5 engineers in designing, developing, and testing SCADA-related embedded products (PCBPLC, iRTU6000, SCANET); delivered 4 major product upgrades within 18 months

- Architected firmware for LPC2378-based Intelligent Remote Terminal Units (iRTU) with multitasking RTOS; reduced field failure rate by 35% through rigorous unit and integration testing
- Developed MODBUS RTU/TCP and proprietary protocols for communication between RTUs and SCADA servers; enabled 24/7 remote monitoring for 30+ pumping stations
- Authored ISO 9001 documentation and test plans; mentored junior engineers on embedded C, hardware debugging, and protocol implementation
- Collaborated with project managers to define technical roadmaps; achieved 100% on-time delivery for all assigned projects.

Sr. Project Engr • EXPLORA INFOTECH LTD • Ahmedabad

Feb '04 – Aug '06

Designed and supervised installation of integrated access control amp; man management systems for government buildings; managed 15-member cross-functional team across 5 gates, 10 blocks

- Integrated RF smart card readers, long-range RFID, boom barriers, and turnstiles via TCP/IP; reduced unauthorized access incidents by 90% at client sites
- Prepared technical specifications, electrical diagrams, and commissioning documentation for 3 large-scale security projects (gt;₹2Cr each)
- Provided tender assistance for government contracts; contributed to winning 2 major bids through detailed system design and cost optimization.

R&D Engr • Instruments Universal • Vadodara, Gujarat, India

Sep '02 – Jan '04

Designed and tested 6 microcontroller-based products (automatic samplers, street light controllers, sequential timers) for textile, plastic molding, and chemical industries

- Converted legacy assembly-language firmware to C, improving code maintainability and reducing debugging time by 50%
- Developed PCB layouts using OrCAD and Protel; achieved first-pass success on 80% of new designs.

Software Programmer • E Cube Solutions Pvt Limited • Vadodara, Gujarat, India

Jul '01 – Aug '02

Programmed 89C51-based access control and time attendance systems; implemented ISO 7816 smart card protocols (T=0, T=1) and Mifare (ISO 14443) contactless interface

- Developed firmware for smart card POS terminals with 10,000-record storage; enabled loyalty program tracking for retail clients
- Collaborated with hardware team to select components and debug prototypes; contributed to 3 product launches within first year.

Skills

Top Skills: General Packet Radio Service (GPRS), NXP, Ethernet, modbus tcp, RS232, LPC, Regression Testing, System Testing, Test Cases, Device Drivers, Assembler, Testing, Computer-Aided Design (CAD), Technical Support, Architectural Design, RTC, Assembly Language, Energy Management, ASTRONOMICAL TIMER, Technical Documentation, Smart Cards, Radio-Frequency Identification (RFID), Internet Protocol Suite (TCP/IP), Telematics, Automotive Electronics, CAN bus, Programmable Logic Controller (PLC), RS485, WiFi, Radio Frequency (RF) Engineering, Industrial Internet of Things (IIoT), System Integration, Global System for Mobile Communications (GSM), Message Oriented Middleware (MOM), Firmware, Electronic Engineering, Arduino IDE, Remote Device Management, Industrial Design, ARM Cortex-M, ESP32 Microcontrollers, Product Design, EasyEDA, Remote Monitoring, New Product Implementations, Data Monitoring, Battery Management Systems, Distributed Teams, Embedded Devices, Remote Terminal Unit (RTU)

Projects

Automatic Street Light Controller & Remote Monitoring | [Link](#)

Automotive Battery Management System (BMS) | [Link](#)

GPS/GSM Vehicle Tracking & Fleet Monitoring System | [Link](#)

Industrial IoT Gateway – MODBUS & MQTT | [Link](#)

Integrated Access Control & Security Management System | [Link](#)

Intelligent RTU / PLC for SCADA & Pumping Stations | [Link](#)

Education

Rosary High School, VADODARA, GUJARAT, India | HIGHER SENCONDARY SCHOOL
SCIENCE

Jun '95 – Mar '97

- **Rosary Higher Secondary School — Science**

- Completed higher secondary education with a Science stream.
- Built a strong foundation in Physics, Chemistry, Mathematics, and core scientific principles.
- Developed analytical thinking, problem-solving, logical reasoning, and technical aptitude.
- Established the academic foundation for pursuing engineering in Electronics & Communication.

Dharmsinh Desai University, Nadiad, Gujarat, India | Bachelor of Engineering, Electronics and
communication
Electronics and communication

Jul '97 – May '01

- **Dharmsinh Desai University, Nadiad, Gujarat — B.E. Electronics & Communication**

- Completed Bachelor of Engineering in Electronics & Communication.
- Developed strong foundations in electronics, communication systems, embedded systems, microcontrollers, digital/analog electronics, and control systems.
- Gained practical exposure to hardware design, circuit development, programming, instrumentation, and system integration.
- Built a strong engineering foundation that supported a career in embedded systems, industrial automation, IoT, and electronic product development.